

Superconducting Magnetic Levitation with a Rotating Double Helix

TZPID Companion Spine Series, Paper C19 of XXVII
A complete mathematical model for helical levitation at human scale

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Abstract

This companion develops the complete mathematical model for a human-scale superconducting magnetic levitation system whose field geometry is a rotating double helix inspired by DNA B-form structure. The source manuscript’s stated thesis: “Superconducting Magnetic Levitation with Rotating Double Helix Daniel Alexander Trawin Superconducting Magnetic Levitation with Rotating Double Helix: Complete Mathematical Model Introduction This analysis develops a complete mathematical framework for a human-scale superconducting magnetic levitation system featuring a rotating double helix geometry inspired by DNA B-form structure. The system achieves stable magnet...” The paper organizes this thesis as a TZPID gold spine: a curated seven-node registry chain (ID3321, ID9399, ID0409, ID1935, ID1780, ID2301, ID1955) with typed proof-obligation roles, dictionary and encyclopedia anchors for every node, a declared dependency order, and stubs in the program’s Isabelle/Wolfram verification pipeline. Within the arc this paper is the movement engine’s flagship apparatus: helicity engineering rendered as a buildable machine.

1 Introduction

If the corpus’s movement physics is right, it should be able to hold something up. This companion accepts the dare at full scale: a complete mathematical model for a human-scale superconducting magnetic levitation system whose distinctive choice is its field geometry—a rotating double helix, proportioned after the DNA B-form, rather than the conventional axisymmetric bearing arrangements of levitation engineering.

The choice is the physics. A double-helix field configuration is helicity made hardware: two linked flux families whose Gauss linking (core Paper VIII’s Lk integral, Paper IV’s helicity invariant) is built in by the winding geometry rather than induced dynamically. Rotation activates the gyromagnetic chain—the rotating helical field is a driven version of the configurations whose state space, helicity bookkeeping, and Elsasser balance Papers IV and C17 typed—and superconductivity supplies the flux pinning and persistent currents that make the linked configuration hold without dissipation. Stability, the perennial obstacle of magnetic levitation (Earnshaw’s theorem forbids static passive equilibria of fixed magnets), is purchased here jointly by the superconductor’s flux pinning and the rotation’s gyroscopic stiffness: the system is engineered inside the loopholes.

The spine below anchors the model’s registry chain. As with the TZPQVS synthesis (C5), this is an engineering companion: its obligations are load capacities, stiffness matrices, and quench margins, and its refinement pass should attach numbers to each. Its value to the arc is demonstrative—a single built apparatus exhibiting linked-flux levitation at the modeled parameters would be the corpus’s most photogenic validation.

2 The concept-anchored spine methodology

The companion series applies a uniform method, stated here so that the paper is self-contained. The source manuscript was published with its mathematics rendered as text rather than as machine-readable LaTeX, so its spine is *concept-anchored*: the thesis is taken from the manuscript’s abstract, and the spine nodes are the canonical-registry entries most strongly matched to the manuscript’s themes, selected by weighted term overlap with foundational nodes ranked first. Each node carries its registry equation, its dictionary definition, its encyclopedia interpretation, and a declared proof-obligation role (Core_Definition, Core_Axiom, or Derived_Theorem_Obligation). The resulting chain is a starting backbone for refinement—a typed scaffold onto which an exact, equation-by-equation crosswalk with the source manuscript can be progressively attached—rather than a claim of completed formalization.

Three properties make the backbone useful even before refinement completes. First, *addressability*: every claim in the companion paper resolves to a registry identifier whose equation, definition, and verification status are independently inspectable, so disagreement can be localized to a node rather than to the paper as a whole. Second, *pipeline coupling*: each spine is paired with an Isabelle focus-theory stub and a Wolfram check stub, so that as nodes are refined their obligations enter the same machine-checked pipeline that certifies the core series’ guardrails. Third, *role discipline*: the obligation roles separate what the spine defines from what it asserts and what it must prove, which keeps the demarcation section of every companion paper honest by construction.

The dependency chain should accordingly be read as a proof-order proposal: upstream nodes supply the vocabulary and axioms that downstream nodes consume, and the refinement pass is free to reorder where the source manuscript’s own logic demands it. Where a node’s registry equation arrives with rendering artifacts inherited from the source extraction (a known cost of text-rendered mathematics), the equation is quoted as carried by the registry, with the dictionary and encyclopedia entries supplying the authoritative reading.

3 The registry chain

The curated chain, in proof order, is

$$\text{ID3321} \rightarrow \text{ID9399} \rightarrow \text{ID0409} \rightarrow \text{ID1935} \\ \rightarrow \text{ID1780} \rightarrow \text{ID2301} \rightarrow \text{ID1955}.$$

We take the nodes in order; each subsection quotes the registry equation as carried by the canonical master, then gives the dictionary definition and the encyclopedia’s interpretive reading.

3.1 ID3321: Magnetic Flux Relation (Core_Definition)

Registry equation:

$$\begin{equation} \Phi_{\text{total}} = \psi_{\text{quantize}} \circ \pi_{\text{propagate}} \\ \circ \eta_{\text{holographic}} \circ \varepsilon_{\text{encode}} \end{equation}$$

Dictionary definition. H\$ transforms discrete sphere points into continuous wavefront configurations through spherical harmonic decomposition.

Encyclopedia reading. The entry preserves the equation exactly as extracted from ‘00_EngineLemma-MakerPrismQuantumTopoHoloNavSystem_formatted2.tex’ for registry alignment and later derivation.

Computational guardrail: MagneticLevitation_ID3321_identity.

Obligation reading. As a Core_Definition this node contributes vocabulary rather than assertion: downstream nodes may use its object freely, but nothing is yet claimed about the world. Its refinement burden is precision—the typed form must capture exactly the object the source manuscript deploys.

3.2 ID9399: Definition of SDAAN (Core_Definition)

Registry equation:

$$\text{SDAAN} = S + \text{cup} S$$

Dictionary definition. Definition of SDAAN — $\text{SDAAN} = S + \text{cup} S$

Encyclopedia reading. Definition of SDAAN is an algebraic definition, written $\text{SDAAN} = S + \text{cup} S$. It is an algebraic definition expressing the quantity directly in terms of the variables on its right-hand side. It is built from S , defining SDAAN. Within the Trawin Zero Point registry it serves as one element of the framework's mathematical narrative.

Computational guardrail: `MagneticLevitation_ID9399_identity`.

Obligation reading. This `Core_Definition` fixes an object the rest of the chain consumes. In proof order it must be discharged first; in refinement it is where a mismatch between the manuscript's usage and the registry's typing would first surface.

3.3 ID0409: The Superconducting Pair Creation Operator (Derived_Theorem_Obligation)

Registry equation:

$$\begin{array}{c} \begin{array}{c} \text{\texttt{\textbackslash begin\{adjustbox\}\{max width=\linewidth,center\} \\\text{\textbackslash displaystyle \hat{H}_{\mathrm{SC}}=\Delta\text{\textbackslash,c_{\uparrow}w}^{\text{\textbackslash dagger}c_{\downarrow}\text{\textbackslash dagger}+\overline{\Delta}\text{\textbackslash,c_{\downarrow}c_{\uparrow}w} \\\text{\textbackslash end\{adjustbox\}}} \end{array} \end{array}$$

Dictionary definition. ID 409 defines the quantum interface between the rotating ferrite core and the superconducting levitation subsystem.

Encyclopedia reading. T: Carrier Map is localized within the manifold and metric dynamics arc of the directory. R: Composable Action preserves the operative relation that the tunnelling barrier operator (`hath texttunnel`) requires. A: Normalized Output fixes the admissible closure condition for the entry. W: THE TUNNELLING BARRIER OPERATOR (`hath texttunnel`) is rendered as a reusable operator-level statement. I: The informational content of the tunnelling barrier operator (`hath texttunnel`) is retained rather than flattened. N: THE TUNNELLING BARRIER OPERATOR (`hath texttunnel`) closes as a distinct normalized TRAWIN object. Interpretive role: This entry specifies the structure within the registry.

Computational guardrail: `MagneticLevitation_ID0409_identity`.

Obligation reading. This node carries proof debt by declaration—`Derived_Theorem_Obligation`—and is therefore where the refinement pass should concentrate first: a discharged derivation here strengthens every downstream node simultaneously.

3.4 ID1935: EID_0924 Magnetic Field Source (Derived_Theorem_Obligation)

Registry equation:

$$\rho_{\text{eff}} = \rho_0 + \alpha N + \beta |\nabla \theta|^2 + \gamma |\psi|^2$$

Dictionary definition. EID 0924 Dialogue [://economics.uwo.ca/math/resources/calculus-multivariable-functions/7-partial-derivatives-constrained-optimization/content/](http://economics.uwo.ca/math/resources/calculus-multivariable-functions/7-partial-derivatives-constrained-optimization/content/)) The system achieves stable magnetic levitation through flux pinning in high-temperature superconductors while the inner rotating helix generates electrical p (ρ_{eff})

Encyclopedia reading. EID 0924 Dialogue [://economics.uwo.ca/math/resources/calculus-multivariable-functions/7-partial-derivatives-constrained-optimization/content/](http://economics.uwo.ca/math/resources/calculus-multivariable-functions/7-partial-derivatives-constrained-optimization/content/)) The system achieves stable magnetic levitation through flux pinning in high-temperature superconductors while the inner rotating helix generates electrical power

through non-contact inductive coupling. Dadao (breakthroughs marks a reasoning pivot toward ACOUSTIC-MAGNETIC LEVITATION FORCE: [://economics.uwo.ca/math/resources/calculus-multivariable-functions/7-partial-derivatives-constrained-optimization/content/](http://economics.uwo.ca/math/resources/calculus-multivariable-functions/7-partial-derivatives-constrained-optimization/content/)) The system achieves stable magnetic levitation through flux pinning in high-temperature superconductors while the inner rotating helix generates electrical power through non-contact inductive coupling. In technical terms, the entry packages the selected formula as the operative carrier, the nearby equations as the progression layer, and th

Computational guardrail: MagneticLevitation_ID1935_identity.

Obligation reading. As a Derived_Theorem_Obligation this node owes a proof: the registry asserts that it follows from the chain's upstream vocabulary and axioms, and the Isabelle focus stub holds the slot where that derivation will be discharged.

3.5 ID1780: MagneticLevitation (Derived_Theorem_Obligation)

Registry equation:

$$P_{1, \mathrm{avg}} \approx \frac{(N_1 B_0 A)^2 \omega}{2L_1}$$

Dictionary definition. Technically, MagneticLevitation is interpreted through the canonical variables in the repaired equation and through the adjacent progression terms retained in the packet.

Encyclopedia reading. Technically, MagneticLevitation is interpreted through the canonical variables in the repaired equation and through the adjacent progression terms retained in the packet.

Computational guardrail: MagneticLevitation_ID1780_structure.

Obligation reading. A derived obligation: the spine claims this follows from what precedes it. Until the formal derivation is supplied, the computational guardrail polices at least the node's internal consistency.

3.6 ID2301: Fractal Frequencies (Derived_Theorem_Obligation)

Registry equation:

$$P_{1, \mathrm{avg}} \approx \frac{(N_1 B_0 A)^2 \omega}{2L_1}$$

Dictionary definition. The selected packet contained a raw prose/code fragment with usable power equations inside it [ID2301]

Encyclopedia reading. The selected packet contained a raw prose/code fragment with usable power equations inside it. The source text supports a clean inductive power-transfer relation for the rotating double-helix magnetic levitation model.

Computational guardrail: MagneticLevitation_ID2301_structure.

Obligation reading. This node carries proof debt by declaration—Derived_Theorem_Obligation—and is therefore where the refinement pass should concentrate first: a discharged derivation here strengthens every downstream node simultaneously.

3.7 ID1955: EID_1049 Magnetic Field Source (Derived_Theorem_Obligation)

Registry equation:

$$\nabla \cdot \vec{J} = -\frac{\partial \rho_e}{\partial t}$$

Dictionary definition. EID_1049 Magnetic Field Source — $\nabla \cdot \vec{J} = -(\partial \rho_e)/(\partial t)$ [ID1955]

Encyclopedia reading. al Effect ————— Double helix charged/magnetic core Helical E and B fields, Eu00d7B drift, ion focusing Vortex flow Mass transport of ions toward center Rotating outer shell

Centrifugal separation, magnetic field shearing Alternating magnetic fields Ion trapping, resonance, wave-particle interaction Varying charge regions Parametric resonance, electric body forces Reionization via shear & impact Sustained plasma-like behavior — ud83dudd2c Key Governing Equations/Effects 1. Maxwell Stress (modifying pressure): \ — ud83eudde9 Conceptual Summary: What You’re Building You’re describing a kind of rotating electro-magnetic vortex engine that acts a An ion mass separator - A re-ionization chamber - An electromagnetic resonance amplifier - A self-focusing ionic capacitor - Possibly even a fluidic-plasma hybrid system Depending on energy input, this may behave like

Computational guardrail: MagneticLevitation_ID1955_identity.

Obligation reading. As a Derived_Theorem_Obligation this node owes a proof: the registry asserts that it follows from the chain’s upstream vocabulary and axioms, and the Isabelle focus stub holds the slot where that derivation will be discharged.

4 Dependency structure and obligations

Table 1 summarizes the spine’s obligation ledger. The chain’s proof order runs from the definitional nodes (vocabulary first) through the derived obligations to the spine’s closure; the refinement pass may interleave additional registry nodes where the source manuscript’s own derivations require them, in which case the table grows monotonically—existing rows are never weakened, only joined.

Table 1: Obligation ledger for the MagneticLevitation spine.

ID	Obligation role	Wolfram check
ID3321	Core_Definition	MagneticLevitation_ID3321_identity
ID9399	Core_Definition	MagneticLevitation_ID9399_identity
ID0409	Derived_Theorem_Obligation	MagneticLevitation_ID0409_identity
ID1935	Derived_Theorem_Obligation	MagneticLevitation_ID1935_identity
ID1780	Derived_Theorem_Obligation	MagneticLevitation_ID1780_structure
ID2301	Derived_Theorem_Obligation	MagneticLevitation_ID2301_structure
ID1955	Derived_Theorem_Obligation	MagneticLevitation_ID1955_identity

5 Crosswalk to the core series and companion corpus

Because all spines draw on one canonical registry, node sharing is measurable and meaningful: a shared identifier means two papers consume literally the same typed object, so verification work transfers between them without translation. Of this spine’s seven nodes, 1 appear in core-series spines and 0 recur elsewhere in the companion corpus. Table 2 records the census.

Two readings follow. Nodes shared with the core series are this spine’s strongest members: their guardrails are already certified in the core pipeline, and the present paper inherits that status. Nodes unique to this spine are its distinctive contribution—and its refinement priority, since no other paper will discharge them. Nodes recurring across companions mark the corpus’s internal seams: the same object consumed by several manuscripts, whose refinements must agree by the duplicate discipline documented in the Einstein Focus companion.

Table 2: Node-sharing census for this spine.

ID	Core-series appearances	Companion-corpus appearances
ID3321	—	—
ID9399	—	—
ID0409	VII (Energy to Matter)	—
ID1935	—	—
ID1780	—	—
ID2301	—	—
ID1955	—	—

6 Position in the TZPID arc

This companion is applied Paper IV: helicity as invariant becomes helicity as blueprint, the Elsasser balance becomes an operating point, and the rotating-metric corrections (negligible at these rim speeds, as Paper IV’s regime table records) are honestly omitted. Its linked-flux geometry also instantiates C16’s toroidal habitat and Paper VIII’s linking integral in copper and niobium rather than notation.

7 Verification pathway

The spine enters the program’s two-engine verification pipeline. The Isabelle focus theory `TZPID_Spine_MagneticLevitation` types the seven obligations over the registry’s declared vocabulary, in the dependency order of Section 3; the Wolfram check stub `MagneticLevitation_checks.wl` carries the per-node computational guardrails listed in Table 1. As with the core series, the division of labor is deliberate: the theorem prover certifies structure (types, roles, dependency soundness), while the computational engine certifies arithmetic and algebraic identities node by node. A check name of the form `..._identity` denotes a per-node identity guardrail generated from the registry equation; refined checks replace these as the equation-level crosswalk matures.

Because the companion spines share the registry with the core series, verification is cumulative: any node here that also appears in a core spine (the chain tables record several) inherits the core guardrail’s status, and any refinement proved here propagates back. The companion series thus functions as the proof pipeline’s breadth dimension—161 distinct registry nodes across the 27 companions—complementing the core series’ depth.

The pipeline’s two-engine division also fixes what failure means at this layer. A failed Wolfram guardrail localizes an arithmetic or algebraic defect in one node’s registered equation: the repair is editorial (correct the registration against the source) or substantive (the source itself errs), and the obligations ledger records which. A failed Isabelle discharge, by contrast, indicts the chain: a type mismatch or unprovable dependency means the proof-order proposal of Section

`refsec:chain` is wrong as stated, and the refinement pass must re-curate. Keeping these failure modes separate is what the stub architecture buys: every companion enters the pipeline with its failure semantics declared in advance, which is the practical meaning of being peer-reviewable by machine.

8 Refinement worksheet

The concept-anchored backbone converts into an equation-level formalization through a bounded, node-by-node worksheet. For each node the worksheet item below states the concrete refinement act; completing all seven closes the spine’s first formalization pass.

1. **ID3321** (`MagneticLevitation_ID3321_identity`): exhibit the source manuscript’s working definition beside the registry equation; reconcile notation; type the object in the focus theory and confirm every downstream node consumes the typed form.
2. **ID9399** (`MagneticLevitation_ID9399_identity`): exhibit the source manuscript’s working definition beside the registry equation; reconcile notation; type the object in the focus theory and confirm every downstream node consumes the typed form.
3. **ID0409** (`MagneticLevitation_ID0409_identity`): reconstruct the source manuscript’s derivation sketch; identify the upstream nodes it actually uses (amending the chain if they differ from the curated order); discharge the proof in the focus theory or record the precise gap.
4. **ID1935** (`MagneticLevitation_ID1935_identity`): reconstruct the source manuscript’s derivation sketch; identify the upstream nodes it actually uses (amending the chain if they differ from the curated order); discharge the proof in the focus theory or record the precise gap.
5. **ID1780** (`MagneticLevitation_ID1780_structure`): reconstruct the source manuscript’s derivation sketch; identify the upstream nodes it actually uses (amending the chain if they differ from the curated order); discharge the proof in the focus theory or record the precise gap.
6. **ID2301** (`MagneticLevitation_ID2301_structure`): reconstruct the source manuscript’s derivation sketch; identify the upstream nodes it actually uses (amending the chain if they differ from the curated order); discharge the proof in the focus theory or record the precise gap.
7. **ID1955** (`MagneticLevitation_ID1955_identity`): reconstruct the source manuscript’s derivation sketch; identify the upstream nodes it actually uses (amending the chain if they differ from the curated order); discharge the proof in the focus theory or record the precise gap.

The worksheet’s order matters: definitional items unblock derived ones, and a failure at any item halts propagation rather than silently weakening downstream claims. Completed items should update the obligations CSV in place, so that the spine’s public ledger always reflects its true refinement state—the same live-ledger discipline the core series applies to its guardrails.

9 Demarcation and refinement targets

As throughout the program, the demarcation line is drawn explicitly. The established layer comprises the standard mathematics and physics that the chain’s nodes quote—definitions, identities, and independently citable results—together with whatever the computational guardrails certify. The framework layer comprises the source manuscript’s interpretive claims and the spine’s structural assertion that its nodes form the stated dependency chain. The model’s mathematics is auditable as mechanics and electrodynamics; every claim beyond it is a bench claim—lift, stiffness, stability envelope—settled by construction and measurement rather than argument.

Refinement targets, in pipeline order: (i) replace the concept-anchored node selection with an equation-level crosswalk against the source manuscript; (ii) promote each `..._identity` guardrail to a content-bearing check; (iii) discharge the typed obligations in the Isabelle focus theory against the program’s shared vocabulary; and (iv) record any node whose refinement fails, since a failed node localizes exactly where the source manuscript and the canonical registry disagree—the information a refinement pass exists to produce.

10 Conclusion

The double-helix levitator is the corpus’s movement physics asked to perform: linking built into windings, gyroscopic stiffness from rotation, persistence from superconductivity. The refinement targets are an engineer’s—tolerances, margins, quench behavior—and the payoff of a working build would be disproportionate: nothing argues for a framework like an object that floats because of it.

References

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